Case 1 — Mr. Samuel Okoro

68 M | 5 days post-ischaemic stroke — Right MCA | Acute Stroke Unit, medically stable | Left hemiparesis (UL + face)

Target: 70%+ | Neurological Physiotherapy

■ PROBLEM LIST

- Left hemiparesis (UL + face)
- Left-sided neglect
- Pusher syndrome
- Poor trunk control
- 2-person assist STS
- Day 5 acute

■ RED FLAGS

• Positive / abnormal finding: Expected to be very low (<50) given the 2-person assist requirement. Score <50 at Day 5–7 predicts functional dependency at discharge and confirms that trunk control rehabilitation must be my number one priority in weeks 1–2.

■ PATHOPHYSIOLOGY

• 2 PATHOPHYSIOLOGY OF WORKING DIAGNOSIS What Is Happening Neurologically Mr.
• Occlusion of the right MCA results in infarction of the corticospinal tract fibres subserving the contralateral (left) body, producing upper motor neuron hemiparesis.
• Neuroplasticity — encompassing synaptic unmasking, axonal sprouting, and cortical map reorganisation — begins within hours of the infarct and is the biological substrate that physiotherapy exploits through high-repetition, task-specific practice to drive functional recovery.
• Typical Signs & Symptoms Explained by Pathophysiology Sign / Symptom Pathophysiological Explanation Left-sided hemiparesis (UL > LL) Right motor cortex / corticospinal tract infarction.
• UL fibres occupy a more lateral and superior position in MCA territory, hence disproportionately affected by MCA occlusion compared with the more medially represented LL fibres (ACA territory).

■ DIFFERENTIAL DIAGNOSES

—

■ OBJECTIVE ASSESSMENTS & HOW TO PERFORM

Scale for Contraversive Pushing (SCP)
What it measures: The severity of pusher syndrome — specifically, the degree to which the patient spontaneously pushes toward the hemiparetic side and actively resists passive correction back to midline. Why I choose it: Pusher syndrome is the primary safety barrier for this pati

Line Bisection Test (Hemispatial Neglect Screen)
What it measures: The presence and severity of left hemispatial neglect — specifically, the patient's ability to allocate attention equally to both halves of extrapersonal space. Why I choose it: Neglect independently predicts functional outcome and fall risk post-stroke. It dire

Trunk Control Test (TCT)
What it measures: Sitting and lying-to-sitting postural control across four tasks — rolling to the weak side, rolling to the strong side, sitting up from lying, and static sitting balance. Scored 0–100; <50 at one week post-stroke strongly predicts dependency at discharge. Why I

■ TREATMENT PLAN

■ Initial Phase — Weeks 1–4 (Acute Inpatient)

■ Core Neurophysio Interventions:

- Early mobilisation (upright positioning and sitting out within 48–72 hours of stroke, medically stable):
- Pusher syndrome treatment via vertical visual reference:
- During all sitting and standing practice I will use a vertical reference (door frame, vertical line on a mirror, or a plumb line) and instruct the patient to align his trunk with it.
- Left neglect management — bottom-up cuing and environmental modification:
- Bottom-up attentional cuing temporarily overcomes neglect-related inattention and, with repetition, begins to drive neuroplastic reorganisation of the right hemisphere's attentional network.

■ Long-Term Plan — Weeks 4–12+ (Rehabilitation Ward / Community)

- Transfer to inpatient stroke rehabilitation ward
- for intensive MDT rehabilitation once meeting progression criteria above.
- Gait re-education:
- Parallel bars → wheeled walking frame → single-point stick, with ongoing focus on symmetric gait pattern, left limb control, and ankle stability (consider AFO if foot drop emerges with spasticity).
- Upper limb rehabilitation:

■ Discharge criteria:

• Safe independent or supervised mobility for home environment, carer competent in all safe handling, community supports confirmed, secondary prevention plan in place.
• Examiner Demonstration — What I Would Show in the Room
• STS facilitation technique:
• Safe hemiplegic shoulder handling during a transfer:
• Demonstrate correct support under the elbow, correct arm positioning in sitting, and clearly narrate what must be avoided (traction on the arm). Show the carer exactly how to assist safely.